

Assignment

1. Prove that the subset $W = \{(x, y, z) \mid x - 3y + 4z = 0\}$ of the vector space $\mathbb{R}^3$ is subspace of $\mathbb{R}^3$

$$\left[\begin{array}{c} 1 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 1 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$\frac{1}{2} = \frac{1}{2} + \frac{0}{2}$$

$$\frac{1}{2} = \frac{1}{2} + \frac{0}{2}$$

$$(1) \vec{p} = \frac{10 \vec{u}_1}{2} = \left[\begin{array}{c} 5 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 5 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 7 \vec{u}_1 = \left[\begin{array}{c} 7 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 7 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 10 \vec{u}_1 = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 10 \vec{u}_1 = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 10 \vec{u}_1 = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 10 \vec{u}_1 = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 10 \vec{u}_1 = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$$(1) \vec{p} = 10 \vec{u}_1 = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] = \left[\begin{array}{c} 10 \\ 0 \\ 0 \end{array} \right] + \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

2. Using the convolution theorem, find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$

$$\mathcal{L}^{-1} \left[\frac{s}{(s^2+a^2)^2} \right] = ?$$

$$\bar{f}(s) = \frac{1}{s^2+a^2}$$

$$\bar{g}(s) = \frac{s}{s^2+a^2}$$

$$\mathcal{L}^{-1} [\bar{f}(s)] = \mathcal{L}^{-1} \left[\frac{1}{s^2+a^2} \right] = \frac{\sin at}{a} = f(t)$$

$$\mathcal{L}^{-1} [\bar{g}(s)] = \mathcal{L}^{-1} \left[\frac{s}{s^2+a^2} \right] = \cos at = g(t)$$

By convolution,

$$\mathcal{L}^{-1} [\bar{f}(s) \bar{g}(s)] = \int_{u=0}^t f(u) g(t-u) du$$

$$\mathcal{L}^{-1} \left[\frac{s}{(s^2+a^2)^2} \right] = \int_{u=0}^t \frac{\sin au}{a} \cdot \cos at du$$

$$= \frac{1}{a} \int_{u=0}^t \sin au \cos (at-au) du$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$= \frac{1}{a} \int_{u=0}^t \frac{1}{2} \{ \sin (au+at-au) - \sin (au-at+au) \} du$$

$$= \frac{1}{2a} \int_{u=0}^t \{ \sin at - \sin (2au - at) \} du$$

$$= \frac{1}{2a} \left[\sin at [u]_{u=0}^t - \left[\frac{-\cos (2au - at)}{2a} \right]_{u=0}^t \right]$$

$$= \frac{1}{2a} \left[\sin at (t - 0) + \frac{1}{2a} (\cos at - \cos (-at)) \right]$$

$$= \frac{1}{2a} \left[t \sin at + \frac{1}{2a} (\cancel{\cos at} - \cancel{\cos at}) \right]$$

$$= \frac{1}{2a} t \sin at$$

$$\therefore \mathcal{L}^{-1} \left[\frac{s}{s^2 + a^2} \right] = \frac{t \sin at}{2a}$$

4. Find the Laplace transform of.

(i) $e^{-4t}(2\cos 6t - 3\sin 5t)$

$$4ii. \mathcal{L} \left[\frac{\cos 2t - \cos 3t}{t} \right] = ?$$

$$f(t) = \cos 2t - \cos 3t$$

$$\mathcal{L}[f(t)] = \mathcal{L}[\cos 2t] - \mathcal{L}[\cos 3t]$$

$$= \frac{s}{s^2 + 2^2} - \frac{s}{s^2 + 3^2} = \bar{f}(s)$$

using this

$$\mathcal{L} \left[\frac{f(t)}{t} \right] = \int_s^\infty \bar{f}(s) ds$$

$$\therefore \mathcal{L} \left[\frac{\cos 2t - \cos 3t}{t} \right] = \int_s^\infty \left(\frac{s}{s^2 + 2^2} - \frac{s}{s^2 + 3^2} \right) ds$$

$$= \frac{1}{2} \left[\log(s^2 + 2^2) - \log(s^2 + 3^2) \right]_s^\infty$$

$$= \frac{1}{2} \left[\log \left(\frac{s^2 + 2^2}{s^2 + 3^2} \right) \right]_s^\infty$$

$$= \frac{1}{2} \left\{ \lim_{s \rightarrow \infty} \log \frac{s^2(1 + 2^2/s^2)}{s^2(1 + 3^2/s^2)} - \log \left(\frac{s^2 + 2^2}{s^2 + 3^2} \right) \right\}$$

$$= \frac{1}{2} \left\{ 0 - \log \frac{s^2 + 2^2}{s^2 + 3^2} \right\}$$

$$= \frac{1}{2} \log \frac{s^2 + 3^2}{s^2 + 2^2}$$

$$= \log \sqrt{\frac{s^2 + 3^2}{s^2 + 2^2}}$$

5. Find $f(9)$ from the data Newton's divided difference formula

x	5	7	11	13	17
F(x)	150	392	1452	2366	5202

x	$f(x)$	1 st D.D	2 nd D.D.	3 rd D.D.
$x_0 = 5$	$f(x_0) = 150$	$f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$ $= \frac{392 - 150}{7 - 5} = \frac{242}{2} = 121$	$f(x_0, x_1, x_2) = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}$ $= \frac{1452 - 392}{11 - 5} = \frac{1060}{6} = 176\frac{2}{3}$	$f(x_0, x_1, x_2, x_3) = \frac{f(x_1, x_2, x_3) - f(x_0, x_1, x_2)}{x_3 - x_0}$ $= \frac{2366 - 1452}{13 - 5} = \frac{914}{8} = 114\frac{1}{2}$
$x_1 = 7$	$f(x_1) = 392$	$f(x_1, x_2) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$ $= \frac{1452 - 392}{11 - 7} = \frac{1060}{4} = 265$	$f(x_1, x_2, x_3) = \frac{f(x_2, x_3) - f(x_1, x_2)}{x_3 - x_1}$ $= \frac{2366 - 1452}{13 - 11} = \frac{914}{2} = 457$	$f(x_1, x_2, x_3, x_4) = \frac{f(x_2, x_3, x_4) - f(x_1, x_2, x_3)}{x_4 - x_1}$ $= \frac{5202 - 2366}{17 - 11} = \frac{2836}{6} = 472\frac{2}{3}$
$x_2 = 11$	$f(x_2) = 1452$	$f(x_2, x_3) = \frac{f(x_3) - f(x_2)}{x_3 - x_2}$ $= \frac{2366 - 1452}{13 - 11} = \frac{914}{2} = 457$	$f(x_2, x_3, x_4) = \frac{f(x_3, x_4) - f(x_2, x_3)}{x_4 - x_2}$ $= \frac{5202 - 2366}{17 - 13} = \frac{2836}{4} = 709$	
$x_3 = 13$	$f(x_3) = 2366$			
$x_4 = 17$	$f(x_4) = 5202$			
4 th D.D				

$$y = f(x) = f(x_0) + (x-x_0)f(x_0, x_1) + (x-x_0)(x-x_1)f(x_0, x_1, x_2) + (x-x_0)(x-x_1)(x-x_2)f(x_0, x_1, x_2, x_3) + (x-x_0)(x-x_1)(x-x_2)(x-x_3)f(x_0, x_1, x_2, x_3, x_4)$$

$$\begin{aligned} y = f(9) &= 150 + (9-5)(121) + (9-5)(9-7)(24) + (9-5)(9-7)(9-11)(1) \\ &\quad + (9-5)(9-7)(9-11)(9-13)(0) \\ &= 150 + (4)(121) + 4(2)(24) + 4(2)(-2)(1) \\ &= 150 + 484 + 192 + (-16) \\ &= \underline{\underline{810}} \end{aligned}$$

6. Use Lagrange's interpolation formula to find y at $x=4$, for the given data.

x	0	1	2	5
y	2	3	12	147

$$\begin{aligned} x_0 &= 0 & x_1 &= 1 & x_2 &= 2 & x_3 &= 5 \\ y_0 &= 2 & y_1 &= 3 & y_2 &= 12 & y_3 &= 147 \end{aligned}$$

$$\begin{aligned} y = f(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \\ &\quad \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3 \end{aligned}$$

$$\begin{aligned} y = f(4) &= \frac{(4-1)(4-2)(4-5)}{(0-1)(0-2)(0-5)} (2) + \frac{(4-0)(4-2)(4-5)}{(1-0)(1-2)(1-5)} (3) + \\ &\quad \frac{(4-0)(4-1)(4-5)}{(2-0)(2-1)(2-5)} (12) + \frac{(4-0)(4-1)(4-2)}{(5-0)(5-1)(5-2)} (147) \end{aligned}$$

$$= \frac{(3)(2)(-1)}{(1)(-2)(-5)} (2) + \frac{(4)(2)(-1)}{(1)(-1)(-4)} (3) + \frac{(4)(3)(-1)}{(2)(1)(-3)} (12) +$$

$$\frac{(4)(3)(2)}{(5)(4)(3)} (147)$$

$$= \frac{3}{5} (2) + \frac{2}{-1} (3) + \frac{2}{1} (12) + \frac{2}{5} (147)$$

$$= 1.2 + (-6) + 24 + 58.8$$

$$= \underline{\underline{78}}$$

7. Using Modified Euler's method find y at $x = 0.1$ given $y' = x^2 + y$ with $y(0) = 1$ $h = 0.05$

$$f(x, y) = x^2 + y \quad x_0 = 0 \quad y_0 = 1 \quad h = 0.05$$

$$x_1 = x_0 + h$$

$$= 0 + 0.05$$

$$f(x_0, y_0) = 0^2 + 1 = 1$$

$$x_1 = 0.05$$

Ist stage

Euler's formula.

$$y_1^{(0)} = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.05 f(0, 1)$$

$$= 1 + 0.05 (0^2 + 1)$$

$$= 1 + 0.05 (1)$$

$$= \underline{\underline{1.05}}$$

Modified Euler's method:

$$\begin{aligned}y_1^{(1)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})] \\&= 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 1.05)] \\&= 1 + 0.025 [(0^2 + 1) + ((0.05)^2 + (1.05))] \\&= 1 + 0.025 [1 + 1.0525] \\&= 1 + 0.05131 = \underline{\underline{1.05131}}\end{aligned}$$

$$\begin{aligned}y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\&= 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 1.0513)] \\&= 1 + 0.025 [1 + ((0.05)^2 + 1.0513)] \\&= 1 + 0.025 [2.0538] \\&= 1 + 0.0513 = \underline{\underline{1.0513}}\end{aligned}$$

$$\therefore y(0.05) = \underline{\underline{1.0513}}$$

IInd stage:

$$y_0 = 1.0513$$

$$x_0 = 0.05$$

$$h = 0.05$$

$$\begin{aligned}x_1 &= x_0 + h \\&= 0.05 + 0.05\end{aligned}$$

$$x_1 = 0.1$$

$$\begin{aligned}f(x_0, y_0) &= (0.05)^2 + 1.0513 \\&= 1.0538\end{aligned}$$

Euler's method.

$$\begin{aligned}y_1^{(0)} &= y_0 + h f(x_0, y_0) \\&= 1.0513 + 0.05 f(0.05, 1.0513) \\&= 1.0513 + 0.05 (1.05138) \\&= 1.10399\end{aligned}$$

Modified Euler's method.

$$\begin{aligned}y_1^{(1)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})] \\&= 1.0513 + \frac{0.05}{2} [f(0.05, 1.0513) + f(0.1, 1.10399)] \\&= 1.0513 + 0.025 [1.0538 + (0.1)^2 + 1.10399] \\&= 1.0513 + 0.025 (2.16779) \\&= 1.10549\end{aligned}$$

$$\begin{aligned}y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\&= 1.0513 + \frac{0.05}{2} [f(0.05, 1.0513) + f(0.1, 1.1054)] \\&= 1.0513 + 0.025 [1.0538 + (0.1)^2 + 1.1054] \\&= 1.0513 + 0.025 (2.1692) \\&= 1.1055\end{aligned}$$

$$\therefore y(0.1) = \underline{\underline{1.105}}$$

8. Apply Milne's predictor-corrector formulae to compute $y(4.5)$ given that $5x \frac{dy}{dx} = 2 - y^2$ and.

x	4.1	4.2	4.3	4.4
y	1.0049	1.0097	1.0143	1.0187

$$y' = \frac{2 - y^2}{5x} \quad h = 0.1$$

x	y	$y' = \frac{2 - y^2}{5x}$
$x_0 = 4.1$	$y_0 = 1.0049$	$y'_0 = \frac{2 - (1.0049)^2}{5(4.1)} = \frac{2 - 1.0098}{20.5} = \frac{0.9902}{20.5}$
$x_1 = 4.2$	$y_1 = 1.0097$	$y'_1 = \frac{2 - (1.0097)^2}{5(4.2)} = \frac{2 - 1.0195}{21} = \frac{0.9805}{21} = 0.04669$
$x_2 = 4.3$	$y_2 = 1.0143$	$y'_2 = \frac{2 - (1.0143)^2}{5(4.3)} = \frac{2 - 1.0288}{21.5} = \frac{0.9711}{21.5} = 0.04517$
$x_3 = 4.4$	$y_3 = 1.0187$	$y'_3 = \frac{2 - (1.0187)^2}{5(4.4)} = \frac{2 - 1.0378}{22} = \frac{0.9622}{22} = 0.04373$
$x_4 = 4.5$	$y_4 = ?$	

$$\begin{aligned}
 y_4^{(p)} &= y_0 + \frac{4h}{3} [2y'_1 - y'_2 + 2y'_3] \\
 &= 1.0049 + \frac{4(0.1)}{3} [2(0.04669) - 0.04517 + 2(0.04373)] \\
 &= 1.0049 + 0.1333 [0.09338 - 0.04517 + 0.08746] \\
 &= 1.0229
 \end{aligned}$$

$$y_4' = \frac{2 - y_4^2}{5x_4} = \frac{2 - (1.0229)^2}{5(4.5)} = \frac{0.95367}{22.5} = 0.04238$$

$$\begin{aligned} y_4^{(c)} &= y_2 + \frac{h}{3} [y_2' + 4y_3' + y_4'] \\ &= 1.0143 + \frac{0.1}{3} [0.04517 + 4(0.04373) + 0.04238] \\ &= 1.0143 + 0.0333 [0.26247] \\ &= 1.0230 \end{aligned}$$

$$y_4' = \frac{2 - y_4^2}{5x_4} = \frac{2 - (1.0230)^2}{5(4.5)} = \frac{0.95347}{22.5} = 0.04237$$

$$\begin{aligned} y_4^{(c)} &= y_2 + \frac{h}{3} [y_2' + 4y_3' + y_4'] \\ &= 1.0143 + \frac{0.1}{3} [0.04517 + 4(0.0437) + 0.04237] \\ &= 1.0143 + 0.0333 [0.26234] \\ &= 1.0230 \end{aligned}$$

$$\therefore y(4.5) = 1.0230$$

9. Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at $(1, 2, 3)$, if

$$\vec{F} = 3x^2\hat{i} + 5xy^2\hat{j} + 5xyz^3\hat{k}$$

$$\Rightarrow \vec{F} = \underbrace{3x^2}_{j_1}\hat{i} + \underbrace{5xy^2}_{j_2}\hat{j} + \underbrace{5xyz^3}_{j_3}\hat{k}$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x} (3x^2) + \frac{\partial}{\partial y} (5xy^2) + \frac{\partial}{\partial z} (5xyz^3)$$

$$= 6x + 10xy + 15xyz^2$$

$$\vec{\nabla} \cdot \vec{F}_{(1,2,3)} = 6(1) + 10(1)(2) + 15(1)(2)(3)^2$$

$$= \underline{\underline{296}}$$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 & 5xy^2 & 5xyz^3 \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y} (5xyz^3) - \frac{\partial}{\partial z} (5xy^2) \right] \hat{i} - \left[\frac{\partial}{\partial x} (5xyz^3) - \frac{\partial}{\partial z} (3x^2) \right] \hat{j} + \left[\frac{\partial}{\partial x} (5xy^2) - \frac{\partial}{\partial y} (3x^2) \right] \hat{k}$$

$$= [5xz^3 - 0] \hat{i} - [5yz^3 - 0] \hat{j} + [5y^2 - 0] \hat{k}$$

$$= 5xz^3 \hat{i} - 5yz^3 \hat{j} + 5y^2 \hat{k}$$

$$\text{curl } \vec{F}_{(1,2,3)} = 5(1)(2)^3 \hat{i} - 5(2)(3)^3 \hat{j} + 5(2)^2 \hat{k}$$

$$= 40 \hat{i} - 270 \hat{j} + 20 \hat{k}$$

10. Show that $\vec{A} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational.

$\Rightarrow$ ~~Re~~ $\vec{A}$

$$\vec{A} = \underbrace{(y^2 - z^2 + 3yz - 2x)}_{f_1} \hat{i} + \underbrace{(3xz + 2xy)}_{f_2} \hat{j} + \underbrace{(3xy - 2xz + 2z)}_{f_3} \hat{k}$$

$$\text{div } \vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= \frac{\partial}{\partial x} (y^2 - z^2 + 3yz - 2x) + \frac{\partial}{\partial y} (3xz + 2xy) + \frac{\partial}{\partial z} (3xy - 2xz + 2z)$$

$$= (-2) + 2x + (-2x + 2)$$

$$= -2 + 2x - 2x + 2$$

$$\text{div } \vec{f} = 0$$

$\therefore$ The given vector is solenoidal.

$$\text{curl } \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 - z^2 + 3yz - 2x & 3xz + 2xy & 3xy - 2xz + 2z \end{vmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (y^2 - z^2 + 3yz - 2x) & (3xz + 2xy) & (3xy - 2xz + 2z) \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y} (3xy - 2xz + 2z) - \frac{\partial}{\partial z} (3xz + 2xy) \right] i - \left[\frac{\partial}{\partial x} (3xy - 2xz + 2z) - \frac{\partial}{\partial z} (y^2 - z^2 + 3yz - 2x) \right] j + \left[\frac{\partial}{\partial x} (3xz + 2xy) - \frac{\partial}{\partial y} (y^2 - z^2 + 3yz - 2x) \right] k$$

$$= \cancel{[3x - 3x]} i - \cancel{[(3y - 2z) - (2z + 3y)]} j$$

$$= [3x - 3x] i - [(3y - 2z) - (2z + 3y)] j + [3x + 2y - (2y + 3x)] k$$

$$= 0i - (3y + 2z - 2z - 3y) j + (3x + 2y - 2y - 3x) k$$

$$= 0i - 0j + 0k$$

$$\therefore \text{curl } \vec{f} = 0$$

$\therefore$ The given vector is irrotational.

3. Prove that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$T(a, b, c) = (3a, a - b, 2a + b + c)$ is a linear transformation.